Phase changes

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for energy transfer during a phase change without a temperature rise.

SUCCESS CRITERIA - I can...

1. Identify a constant-temperature phase-change interval.
2. Use Q=mL for that interval.
3. Separate warming energy from phase-change energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “latent heat” to explain your model or result. Sentence starters are optional.

phase change: a change between states such as solid and liquid

latent heat: energy per mass needed for a specified phase change

temperature: a measure of thermal state; it is not an energy amount

01 Bring back an earlier idea

Two objects at 280 K have different masses and materials. Must their internal energies be equal?
Give two reasons.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: Q = mL; Q = mcΔT.
Use mL during a phase change at constant temperature, and mcΔT outside that interval. L is latent heat.

Worked reasoning: Melting 2 kg of a substance with latent heat 100 J/kg requires 200 J. During ideal melting at fixed pressure, added energy changes phase without raising temperature.

Your turn: Sketch temperature versus added energy through warming and melting.
Mark the plateau.

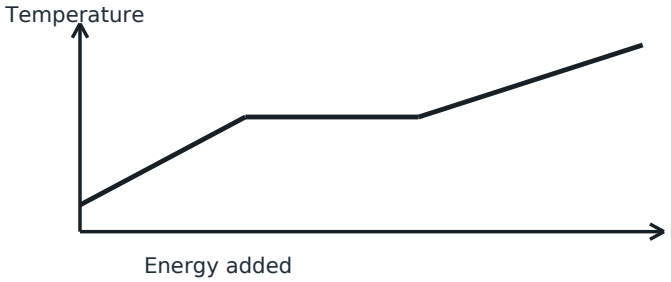
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

On the heating curve, identify the flat interval.
Explain why added energy does not raise temperature during that interval.

Sentence starter: Under the assumption …, the evidence … supports … because … .



The horizontal axis shows energy added. The vertical axis shows temperature.

Marzano check - circle: ① ② ③

04 Solve and justify

Melt .10 kg of ice already at 0°C.
Use L_f=334, 000 J/kg. Then warm the resulting water to 20°C with c=4200 J/(kg·°C).
Find each energy and total.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student uses mcΔT for a melting interval with ΔT=0 and concludes no energy is required.
Explain the missing term.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

06 Choose a method

With right positive, a spring has k=200 N/m and displacement x=-0.06 m.
Find signed spring force and direction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

07 Exit check

A .20 kg liquid releases 45, 000 J while condensing at constant temperature.
Find L and state the sign of Q for the sample.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

A 0.2 kg ice sample at 0°C melts and the water warms to 10°C.
Use L_f=334000 J/kg and c_{water}=4200 J/(kg·°C).
Find total heat.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③